

Określ dziedzinę funkcji $f(x)$, następnie wyznacz jej pochodną $f'(x)$ i określ dziedzinę pochodnej. Oblicz $f'(-1)$ i $f'(2)$.

a) $f(x) = x^5 - 2x^4 + x + 1$

b) $f(x) = \sqrt{6}x + x^7 - \frac{1}{8}x^8$

c) $f(x) = (2x^2 - 3)(4x^2 + 3x)$

d) $f(x) = (x-2)^2(1-x^2)$

e) $f(x) = \frac{3x^2+7x}{6x+1}$

f) $f(x) = \frac{x}{x^5-1}$

g) $f(x) = \frac{3}{x^3-2} - \frac{1}{4x^4}$

a) $f(x) = x^5 - 2x^4 + x + 1, x \in \mathbb{R}$

$f'(x) = 5x^4 - 8x^3 + 1, x \in \mathbb{R}$

$f'(-1) = 5(-1)^4 - 8(-1)^3 + 1 = 5 + 8 + 1 = 14$

$f'(2) = 5 \cdot 2^4 - 8 \cdot 2^3 + 1 = 80 - 64 + 1 = 17$

b) $f(x) = \sqrt{6}x + x^7 - \frac{1}{8}x^8, x \in \mathbb{R}$

$f'(x) = \sqrt{6} + 7x^6 - x^7, x \in \mathbb{R}$

$f'(-1) = \sqrt{6} + 7(-1)^6 - (-1)^7 = \sqrt{6} + 8$

$f'(2) = \sqrt{6} + 7 \cdot 2^6 - 2^7 = \sqrt{6} + 320$

c) $f(x) = (2x^2 - 3)(4x^2 + 3x) = 8x^4 + 6x^3 - 12x^2 - 9x, x \in \mathbb{R}$

$f'(x) = 32x^3 + 18x^2 - 24x - 9, x \in \mathbb{R}$

$f'(-1) = 32(-1)^3 + 18(-1)^2 - 24(-1) - 9 = -32 + 18 + 24 - 9 = 1$

$f'(2) = 32 \cdot 2^3 + 18 \cdot 2^2 - 24 \cdot 2 - 9 = 256 + 72 - 48 - 9 = 271$

$\left[\frac{f(x)}{g(x)} \right]' = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{[g(x)]^2}$

f) $f(x) = \frac{x}{x^5-1}, D_f: \frac{x^5-1 \neq 0}{x \neq 1} \Rightarrow x \in \mathbb{R} \setminus \{1\}$

$f'(x) = \frac{1(x^5-1) - x \cdot 5x^4}{(x^5-1)^2} = \frac{-1-4x^5}{(x^5-1)^2}, D_f: \mathbb{R} \setminus \{1\}$

$f'(-1) = \frac{-1-4(-1)^5}{((-1)^5-1)^2} = \frac{-1+4}{(-1-1)^2} = \frac{3}{4}$

$f'(2) = \frac{-1-4 \cdot 2^5}{(2^5-1)^2} = \frac{-1-4 \cdot 32}{(32-1)^2} = \frac{-129}{961}$

d) $f(x) = (x-2)^2(1-x^2) = (x^2-4x+4)(1-x^2) = x^2-x^4-4x+4x^3+4-4x^2 = -x^4+4x^3-3x^2-4x+4, x \in \mathbb{R}$

$f'(x) = -4x^3+12x^2-6x-4, x \in \mathbb{R}$

$f'(-1) = -4(-1)^3+12(-1)^2-6(-1)-4 = 4+12+6-4 = 18$

$f'(2) = -4 \cdot 2^3+12 \cdot 2^2-6 \cdot 2-4 = -32+48-12-4 = 0$

e) $f(x) = \frac{3x^2+7x}{6x+1}, D_f: \frac{6x+1 \neq 0}{x \neq -\frac{1}{6}} \Rightarrow x \in \mathbb{R} \setminus \{-\frac{1}{6}\}$

$f'(x) = \frac{(6x+7)(6x+1) - 6(3x^2+7x)}{(6x+1)^2} = \frac{36x^2+6x+42x+7-18x^2-42x}{(6x+1)^2} = \frac{18x^2+6x+7}{(6x+1)^2}, D_f: \mathbb{R} \setminus \{-\frac{1}{6}\}$

$f'(-1) = \frac{18(-1)^2+6(-1)+7}{(6(-1)+1)^2} = \frac{18-6+7}{(-6+1)^2} = \frac{19}{25}, f'(2) = \frac{18 \cdot 2^2+6 \cdot 2+7}{(6 \cdot 2+1)^2} = \frac{18 \cdot 4+12+7}{13^2} = \frac{91}{169} = \frac{7}{13}$

g) $f(x) = \frac{3}{x^3-2} - \frac{1}{4x^4}, D_f: \frac{x^3-2 \neq 0 \wedge 4x^4 \neq 0}{x \neq \sqrt[3]{2} \quad x \neq 0}$

$f'(x) = \frac{0 \cdot (x^3-2) - 3(3x^2)}{(x^3-2)^2} - \frac{0 \cdot 4x^4 - 1 \cdot 16x^3}{(4x^4)^2} = \frac{-9x^2}{(x^3-2)^2} + \frac{1}{x^5}$

$D_{f'} = D_f$

$f'(-1) = \frac{-9(-1)^2}{((-1)^3-2)^2} + \frac{1}{(-1)^5} = \frac{-9 \cdot 1}{(-3)^2} + -1 = -2$

$f'(2) = \frac{-9 \cdot 2^2}{(2^3-2)^2} + \frac{1}{2^5} = \frac{-9 \cdot 4}{(8-2)^2} + \frac{1}{32} = \frac{-36}{36} + \frac{1}{32} = -\frac{31}{32}$